

Dynamic Modeling of Large Data Centers Integrated Electric Grid With Impact Analysis and Oscillatory Behavior Assessment

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Abstract—This paper presents an integrated modeling framework that captures both the electrical behavior of data centers and the distinctive operational features of interconnected power grids. A detailed data center model is developed, incorporating server clusters, power electronic interfaces, and standby generation, with particular focus on load dynamics during demand response events, backup power transitions, and power quality disturbances. The model is coupled with an electrical circuit representation of variable IT loads, capturing the dynamic characteristics of AI training workloads. Using this integrated framework, the impact of data center operations on grid performance is evaluated, including voltage variations, loss-of-load events, and fault conditions. In addition, grid-level oscillations associated with high penetration of data center loads are examined. Results indicate that rapid variations in data center load can: (a) shift existing and/or develop new oscillatory modes, (b) increase harmonic distortion, (c) reduce system damping (up to 52%), and (d) induce frequency deviations with a maximum of approximately 0.36 Hz . These findings highlight the potential impact of large-scale data center dynamics on grid stability.

Index Terms—EMT modeling, Data center modeling, Information Technology (IT) load modeling, Data center Uninterruptible Power System (UPS) modeling, Data center modeling for EMT studies.

I. INTRODUCTION

THE rapid growth of large-scale data centers has significantly increased electricity demand and introduced new operational challenges for modern power systems. Recent studies indicate that large-scale computing facilities can reach hundreds of megawatts or even gigawatt-scale loads, making them comparable to large industrial consumers in power systems [1]. Consequently, their integration into transmission and distribution networks requires careful assessment of their electrical characteristics and system-level impacts.

Unlike traditional industrial loads, data centers are dominated by power-electronic equipment such as switched-mode power supplies, uninterruptible power supplies (UPS), variable-speed drives, and inverter-based cooling systems. These converter-based components introduce nonlinear behavior, harmonic distortion, and dynamic load variability in the power system [2]. In addition, the electrical demand of data centers can fluctuate significantly depending on server

utilization, workload patterns, and cooling requirements. Variations in server workload may change power consumption by nearly 100%, resulting in rapid changes in load demand that can influence grid stability and power quality [3]. At the bulk power system level, the increasing penetration of large data center loads introduces several operational and planning challenges as these facilities often connect at transmission voltage levels and may exhibit rapid ramping characteristics due to power electronic controls and computational workload cycles. Recent grid reliability assessments have highlighted that emerging large loads, including hyperscale data centers, can create stability risks related to frequency response, rotor angle stability, voltage stability, and converter-driven interactions [4].

Furthermore, most IT equipment in modern data centers relies on switched-mode power supplies and other power-electronic interfaces, causing the load to behave as a nonlinear source of current distortion. These nonlinear characteristics introduce harmonic currents into the electrical network, which may lead to increased thermal stress in transformers, additional power losses, and voltage waveform distortion in both distribution and transmission systems [5], [6]. With the widespread adoption of high-frequency switching converters and power factor correction circuits, harmonic components and interharmonics may propagate through the network and interact with other power electronic devices [7].

In addition to harmonic distortion, dynamic interactions between converter-based loads and the power system can influence electromechanical oscillations. As data centers increasingly rely on power electronics, their aggregated behavior may affect low-frequency oscillation modes of interconnected power systems. The interaction between converter control systems, UPS architectures, and grid dynamics may either damp or amplify oscillatory modes depending on the control strategies and system conditions. Although extensive research has investigated oscillation phenomena caused by inverter-based generation [8], [9], relatively limited work has focused on the impact of renewable energy and large computational loads on electromechanical oscillation behavior [10].

Modern data centers also employ complex internal electrical infrastructures including utility transformers, switchgear, automatic transfer switches, UPS systems, standby generators, and power distribution units. During grid disturbances such as transient faults or voltage sags, UPS systems temporarily support critical loads while backup generators start up, preventing service interruptions [11], [12]. However, interactions

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between these power electronic devices and the external grid may introduce dynamic phenomena that are not yet fully understood. Also, data centers can potentially participate in demand response programs, load shifting strategies, and frequency regulation services [13], [14].

This paper investigates the impact of large-scale data center loads on power system dynamics, with a focus on low-frequency oscillation modes, harmonic distortion, and grid integration challenges. The work is based on our earlier work in the direction for analyzing grid stability and mitigation with power electronic based energy sources [15]. The study first reviews the electrical characteristics and internal infrastructure of modern data centers. Next, the dynamic interactions between data center loads and power system oscillatory modes are analyzed. Finally, the paper discusses key operational and planning challenges associated with integrating large computational loads into modern power systems. The main contributions of this work are summarized as follows:

- Development of a grid-integrated electromagnetic transient (EMT) modeling framework for large-scale data centers, including dynamic models of key power electronic components such as PSUs, PDUs, and UPS systems including AC-DC, DC-DC, and DC-AC converters), along with analysis of UPS system operational modes.
- Investigation of disturbance-driven dynamic responses of large data center loads under multiple operating scenarios, including temporary grid voltage drops, grid-to-UPS transitions, and UPS-to-grid reconnection events.
- Evaluation of system-level impacts of large-scale data center loads with bulk power grid, including frequency deviation analysis, settling characteristics, electromechanical oscillation modes, damping ratios, and harmonic distortion.
- Comparative assessment of constant data center loads and AI training workload profiles to identify how rapidly varying computational loads influence modal characteristics, oscillation damping, and harmonic distortion in interconnected power systems.

The remaining sections of this paper are organized as follows. Section II discusses the grid integrated data center EMT modeling framework. Section III describes the test system and test case scenarios. Section III-A showcases the simulation results, which includes discussion in Section III-B. Then, the conclusions are presented in Section IV.

II. GRID INTEGRATED DATA CENTER MODELING FRAMEWORK

Fig. 1 shows the one-line diagram of the dynamic data center model. The data center includes a Power Supply Unit (PSU), Power Distribution Unit (PDU), Uninterruptible Power Supply (UPS), Cooling Load, and IT loads, including IT racks. It connects to the electric grid through a 480 V bus, which is stepped up by a medium-voltage transformer (480 V/13 kV). The transmission system and generation station are represented as an additional voltage step-up to 230 kV and connection to the generation station via a transmission line. The generation station consists of a station step-down transmission and a dynamic, full-order generator model with excitation system

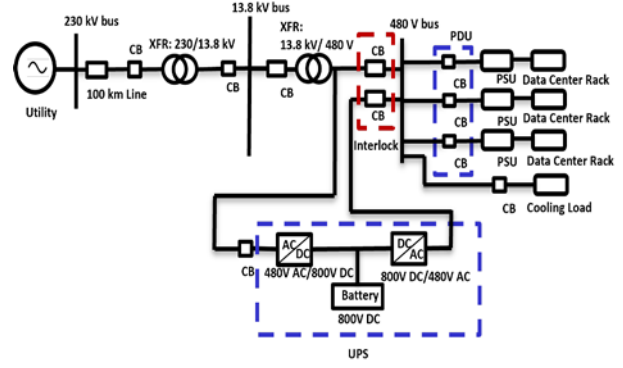


Fig. 1. One line diagram of the dynamic data center model.

and governor. Expanding on the elements shown in Fig. 1, the data center's power electronics can be represented as a Power Supply Unit (PSU) comprising a Power Factor Converter (PFC) and an isolated DC-DC converter. Additionally, a UPS consisting of double-conversion (AC/DC and DC/AC) units is modeled as backup support. The present study primarily focuses on data center load dynamics and therefore cooling loads are not considered.

A. Modeling of PSU

There are two main parts in PSU a) 480 V AC to 480 V DC converter, and b) a DC-DC isolated converter, 480 V DC to 48 V DC, for the IT load. For the 480 V AC-to-480 V DC conversion, the output-voltage regulation loop first generates I_{lref} to determine the duty cycle. This duty cycle, d , is essential for generating the PWM signal required for the converter's control to maintain the desired DC voltage. The associated equations are given below:

$$I_{lref} = K_P(V_{ref} - V_{meas}) + \dots \dots \frac{K_I}{T} \int_0^t (V_{ref} - V_{meas}) dt \quad (1)$$

$$d = K_{P1}(I_{lref} - I_l) + \frac{K_{I1}}{T} \int_0^t (I_{lref} - I_{lmeas}) dt \quad (2)$$

where I_{lref} is the PFC inductor reference current, V_{ref} is the desired reference voltage for PFC, K_P, K_I , are the PID parameters. V_{meas} , & I_{lmeas} are measured PFC voltage and measured PFC inductor current, respectively.

The DC-DC isolated converter also uses a voltage regulation loop, like the AC-DC converter, to sequentially generate the diode current reference and the duty cycle to create a PWM signal for converter control, except it uses a linear transformer to isolate the primary and secondary sides.

B. Modeling of Data Center load

For IT load modeling, a variable resistance is represented as the data center load at the PSU 48 V terminal. The resistance value, R , corresponding to the desired IT load power is obtained as,

$$R = \frac{V_{dc}^2}{P_{IT}} \quad (3)$$

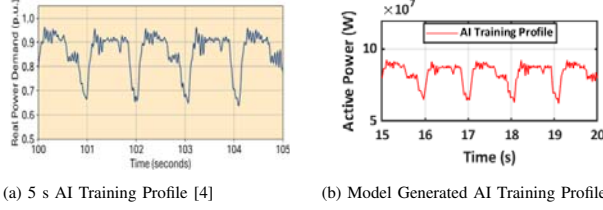


Fig. 2. AI training data center profile.

where V_{dc} represents the DC bus voltage and P_{IT} denotes the IT load power. For a specified P_{IT} value, the equivalent resistance can be calculated directly from (3), allowing the desired data center load profile to be derived.

To represent large-scale data center operation, the PSU-load subsystem is scaled using an equivalent parallel aggregation of multiple units. This scaling approach preserves the electrical behavior of individual racks while enabling representation of high aggregated load levels. Fig. 2 illustrates the AI training workload used in this study. Fig. 2a shows the reference 5 s AI training power profile reported in [4], while Fig. 2b presents the corresponding model-generated load profile using an electric circuit model. The generated profile closely follows the dynamic characteristics of the reference workload and is used to represent rapidly varying AI computational demand.

C. UPS Modeling and Modes of Operation in Data Centers

Uninterruptible Power Supply (UPS) systems constitute a fundamental component of modern data centers, ensuring high availability and power quality for mission-critical information technology (IT) loads. UPS operating modes influence energy efficiency, power conditioning capability, and risk of exposure to voltage or frequency disturbances. Fig. 3 illustrates UPS modes of operation.

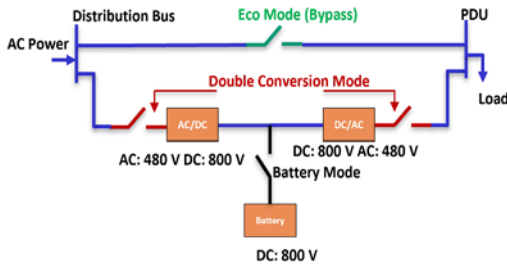


Fig. 3. Modes of operation of UPS.

UPS modeling can be categorized into three major parts. a) AC-DC Conversion model using a 480 V AC to 800 V DC converter, b) 800 V DC battery model, and c) DC-AC Conversion model with a controller for 800 V DC to 480 V AC conversion.

For the AC-DC converter (Rectifier), first I_{Lref} is generated, which is used to generate the duty cycle, D , for the PWM signal. The PWM signal is then used for the converter control to regulate the desired dc voltage. The associated equations are given below:

$$I_{Lref} = K_{P2}(V_{dcref} - V_{dcmeas}) + \dots \dots \frac{K_{I2}}{T} \int_0^t (V_{dcref} - V_{dcmeas})dt \quad (4)$$

$$D = K_{P3}(I_{Lref} - I_L) + \frac{K_{I3}}{T} \int_0^t (I_{Lref} - I_{Lmeas})dt \quad (5)$$

where I_{Lref} is the rectifier inductor reference current, V_{dcref} is the desired DC reference voltage for rectifier, all K_P, K_I are the PID parameters. V_{dcmeas} , & I_{Lmeas} are measured rectifier voltage & measured rectifier inductor current.

The 800 V DC battery is associated with the inverter to regulate the desired AC voltage as well as the required power when needed. With a desired AC voltage magnitude M , desired angle θ and desired frequency f , the input frequency which can be presented as $f(k)$, is first converted into angular frequency,

$$\omega(k) = 2\pi f(k). \quad (6)$$

A discrete-time integrator with sampling period T_s computes the accumulated phase angle,

$$\theta_i(k) = \theta(k-1) + \omega(k)T_s. \quad (7)$$

In order to avoid angle divergence from one electrical cycle, mod operation is used.

$$\theta_m(k) = \text{mod}(\theta(k), 2\pi) \quad (8)$$

The input angle $\phi(k)$ is converted from degrees to radians as,

$$\phi_i(k) = \phi(k) \frac{\pi}{180^\circ}. \quad (9)$$

The three-phase angle displacement is calculated as following:

$$\Delta\phi_{abc} = \begin{bmatrix} 0 \\ -\frac{2\pi}{3} \\ +\frac{2\pi}{3} \end{bmatrix}. \quad (10)$$

The instantaneous phase of each phase is then obtained by the following equation:

$$\theta_{abc}(k) = \theta_m(k) + \phi_i(k) + \Delta\phi_{abc}. \quad (11)$$

Finally, for a given desired voltage magnitude input $M(k)$, the generated three-phase voltages can be expressed as,

$$u_{abcref} = M(k) \sin(\theta_{abc}(k)). \quad (12)$$

This generated u_{abcref} is used for controlling the average model three-level neutral-point-clamped (NPC) converter (inverter).

III. TEST SYSTEM, SIMULATION RESULTS AND DISCUSSIONS

The study is conducted using the standard Kundur two-Area power system [16] shown in Fig. 4. The two Areas are interconnected through long transmission lines and supply aggregated loads for Area-1 at bus-7 and for Area-2 at bus-9. For this study, all Area-1 measurements are from bus-7, and all Area-2 measurements are from bus-9. The Area-1 load is represented as 867 MW conventional load combined with a 100 MW data center load connected through the distribution interface shown in Fig. 1. Area-2 load comprises the original aggregated load of 1767 MW.

To evaluate the dynamic interactions between the power system and the data center load, three disturbance scenarios are considered:

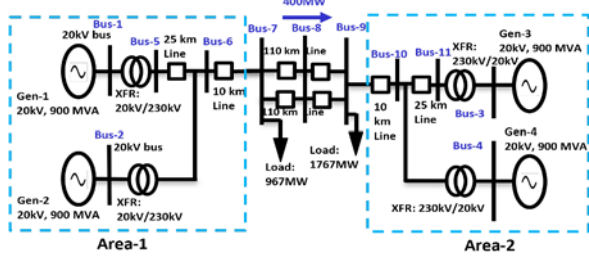


Fig. 4. One-line diagram of the Kundur Two-Area System.

- **Temporary Voltage Sag:** A short-duration voltage sag of 0.75 pu is applied for three cycles at the middle of the tie line at the grid side at $t = 15$ s for a constant data center load study and at $t = 16$ s for an AI training data center load profile study. During this event, the data center remains connected to the grid, and no grid-to-UPS switching occurs.
- **Grid-to-UPS Transition with Load Loss:** A three-phase fault is applied for three cycles at the Area-1 bus on the grid side at $t = 30$ s, causing the grid voltage to temporarily collapse to zero. The data center immediately transitions to UPS operation. As a result, the grid experiences a temporary loss of the 100 MW data center load until reconnection.
- **UPS-to-Grid Reconnection:** At $t = 50$ s, the data center reconnects from UPS operation back to the utility grid.

These disturbance scenarios allow assessment of the dynamic response of the grid–data center interaction under voltage disturbances, load disconnection, and reconnection events, with a constant and an AI-trained data center load profile. The test system was modeled in MATLAB/Simulink (50 μ s discrete time step).

A. Simulation Results

For the simulation demonstration, two data center load types are used. The first is a constant load, and the second is an AI training profile mentioned in [4]. For this study frequency response at various points are evaluated.

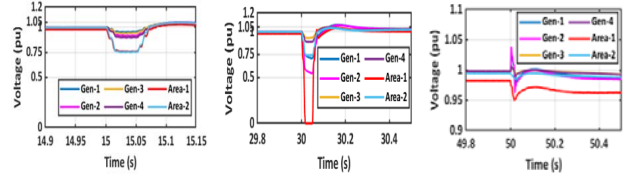
1) Constant Data Center Load

The dynamic responses of the system under constant data center load and disturbance scenarios described in Section III are illustrated in Figs. 5–7, which show the variations in grid voltage and system frequency at Areas 1 and 2 during the considered disturbance events, with the data center load held constant (100 MW). The corresponding frequency deviation characteristics and settling times are summarized in Table I, while the modal characteristics and harmonic distortion associated with these events are summarized in Table II. The percentage change in system frequency due to a disturbance is calculated by summing the maximum rise and drop of the frequency deviation and normalizing it to the nominal system frequency f_{nom} . This is expressed as:

$$\Delta f\% = \frac{f_{rise} + f_{drop}}{f_{nom}} \times 100\%, \quad (13)$$

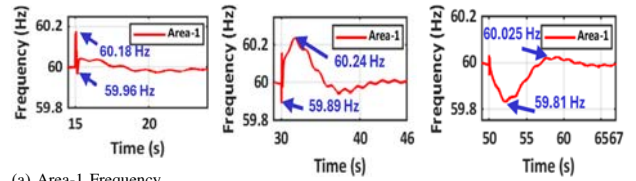
where f_{rise} and f_{drop} are the peak positive and negative deviations of frequency, respectively, and f_{nom} is the nominal system frequency (60Hz).

Please note that constant data center load means a full scale dynamic model of the data center with constant AI training profile.



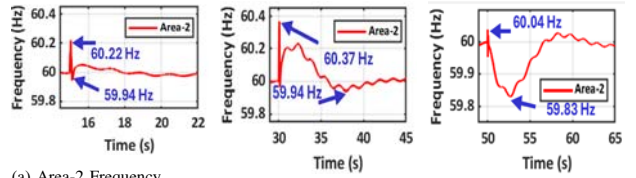
(a) Grid Voltage (Temp. Voltage Drop in Grid) (b) Grid Voltage (Three-Phase Fault in Grid) (c) Grid Voltage (UPS to Grid Transition)

Fig. 5. Grid Voltage with constant data center load.



(a) Area-1 Frequency (Temp. Voltage Drop in Grid) (b) Area-1 Frequency (Three-Phase Fault in Grid) (c) Area-1 Frequency (UPS to Grid Transition)

Fig. 6. Area-1 (Bus 7) Frequency with constant data center load.



(a) Area-2 Frequency (Temp. Voltage Drop in Grid) (b) Area-2 Frequency (Three-Phase Fault in Grid) (c) Area-2 Frequency (UPS to Grid Transition)

Fig. 7. Area-2 (Bus 9) Frequency with constant data center load.

2) AI Training Data Center Load Profile

The dynamic responses of the system under the AI training data center load profile are illustrated in Figs. 8–10. These figures present the variations of grid voltage and system frequency at Areas 1 and 2 under the considered disturbance events, with the AI training data center load profile. The corresponding frequency deviation characteristics and settling times are summarized in Table I, while the modal characteristics and harmonic distortion under these scenarios are summarized in Table II.

B. Discussions

1) Frequency Dynamic Response Under Disturbance Events

The frequency deviation and settling characteristics summarized in Table I highlight several important observations regarding the dynamic impact of data center load disturbances on the power system.

- **Maximum frequency deviation:** Area-2 consistently demonstrates the largest frequency deviation across all

TABLE I
Frequency dynamic characteristics under different data center load profiles and event scenarios (percentage change of nominal frequency 60 Hz)

Data Center Load	Parameters	Case		
		Temporary Voltage Drop at Grid Event	Grid to UPS + Load Loss Event	UPS to Grid Event
Constant Data Center Load	Area-1: Frequency Deviation (R-D)	0.18–0.04 Hz (0.37%)	0.24–0.11 Hz (0.58%)	0.025–0.19 Hz (0.36%)
	Area-1: Settling Time	5 s	11 s	12 s
	Area-2: Frequency Deviation (R-D)	0.22–0.06 Hz (0.47%)	0.36–0.07 Hz (0.72%)	0.04–0.14 Hz (0.30%)
	Area-2: Settling Time	4.5 s	10 s	12 s
AI Training Data Center Profile	Area-1: Frequency Deviation (R-D)	0.17–0.04 Hz (0.35%)	0.22–0.12 Hz (0.57%)	0.04–0.14 Hz (0.30%)
	Area-1: Settling Time	4 s	10 s	7 s
	Area-2: Frequency Deviation (R-D)	0.22–0.06 Hz (0.47%)	0.36–0.07 Hz (0.72%)	0.04–0.14 Hz (0.30%)
	Area-2: Settling Time	4 s	10 s	7 s

Note: R $\Rightarrow$ Rise and D $\Rightarrow$ Drop.

TABLE II
Modal characteristics and harmonic distortion under different data center load profiles and event scenarios

Parameters	Load Condition	Events			
		Standard Kundur System with Ring Down	Temporary Voltage Drop at Grid	Grid to UPS + Load Loss	UPS to Grid Transition
Mode (Hz)	Constant Data Center	0.65, 1.13, 1.18	0.64, 1.11, 1.15	0.63, 1.1, 1.27	0.64, 1.11, 1.29
	AI Training Profile	(Spot Load)	0.64, 0.97, 1.13, 2.3	0.63, 1.1, 1.27	0.65, 1.01, 1.83, 3.24
Damping Ratio (% ζ)	Constant Data Center	2.3, 10.85, 8.32	2.2, 9.88, 8.34	2.4, 10.7, 0.6	1.9, 9.94, 0.55
	AI Training Profile	(Spot Load)	1.41, 2.1, 8.46, 2.65	1.95, 10.81, 1.22	1.1, 3, 3.01, 3.25
THD (%)	Constant Data Center	0.7	2.2	3.58	3.38
	AI Training Profile	(Spot Load)	2	3.07	2.53

Note: indicates mode shifting, and indicates newly introduced modes.

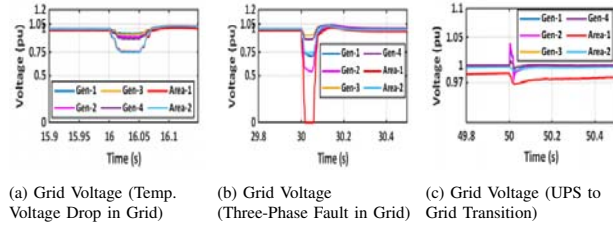


Fig. 8. Grid Voltage with AI Training Data Center Load.

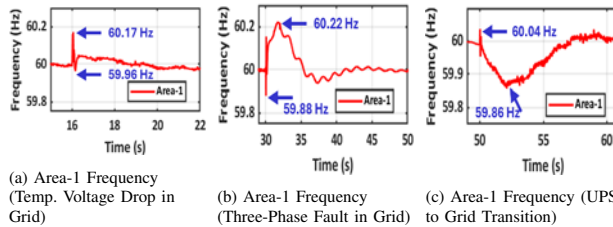


Fig. 9. Area-1 (Bus 7) Frequency with AI Training Data Center Load Profile.

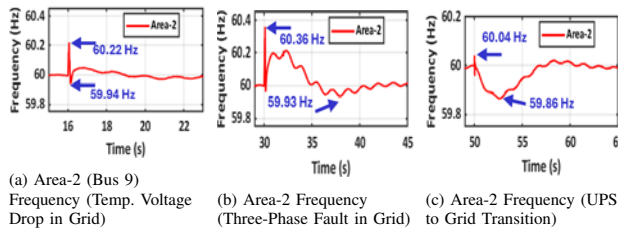


Fig. 10. Area-2 (Bus 9) Frequency with AI Training Data Center Load Profile.

scenarios. During the Grid-to-UPS transition, a load-loss event results in a frequency deviation of approximately 0.36 Hz. These findings suggest that Area-2 undergoes the most significant transient response during disturbances.

- **Most severe disturbance event:** Among the three disturbance scenarios, the Grid-to-UPS transition with load loss produces the largest frequency excursions across the system, indicating that the sudden disconnection of the data center load, along with the Grid-to-UPS transition, introduces the most significant transient imbalance.
- **Settling time characteristics:** The longest system recovery times are also observed during the Grid-to-UPS transition following a load-loss event, with system stabilization occurring in approximately 10–11s.

Overall, the Grid-to-UPS transition with load loss represents the disturbance condition that produces the most pronounced system frequency dynamics, with Area-2 experiencing the largest deviations among the studied locations.

2) Oscillation Modes and Harmonic Characteristics

The modal characteristics and harmonic distortion summarized in Table II provide insight into the oscillatory behavior and power-quality impacts associated with data center load integration.

- **Baseline modes of oscillation:** The third column in Table II provides oscillation modes of the Standard Kundur system with a ring down event. During this event, a fault (ring down event) is initiated and the oscillation modes are evaluated. For this a spot $P - Q$ load is used instead of the dynamic model of the data-center and data center load. The modes of oscillation are being seen (including 0.64 Hz, 1.11 Hz and 1.15 Hz).

- Shift in dominant oscillation modes: Compared to the original two area model without data center load (named as the baseline Kundur system), shifts in oscillation frequencies are observed after integrating the data center load. The dominant inter-area mode near 0.65 Hz varies between approximately 0.63 Hz and 0.65 Hz across disturbance scenarios. In addition, several existing modes above 1 Hz shifts (such as 1.11 Hz shift to approximately 0.97 Hz and 1.01 Hz , 1.15 Hz shifts to 1.13 Hz , 1.29 Hz shifts to 1.83 Hz) as shown in Table II, due to the effect of converter-based data center load dynamic characteristics with varying grid conditions.
- Emergence of additional oscillation modes: Additional oscillation modes appear when the system is exposed to the AI training data center load profile under disturbances. Higher-frequency oscillations at 2.3 Hz and 3.24 Hz appear during disturbances, which is associated with interactions among converter-based data center equipment, disturbance transitions, and the rapidly varying AI training load pattern.
- Reduction in damping of the system: Under the data center load condition, particularly under the AI training load profile, a noticeable reduction in the damping of the dominant inter-area oscillation mode has been observed. For example, the damping ratio of the 0.65 Hz inter-area mode reduces from 2.3% to about 1.1% (approximately 52%) under the considered disturbance scenarios.
- Increase of Harmonic distortion: The presence of converter-based data center loads introduces harmonics as summarized in Table II compared with the baseline Kundur system due to the power-electronic interfaces associated with data center loads.

These observations indicate that while the dominant electromechanical oscillation mode of the system remains largely preserved, the integration of data center loads introduces shifts in oscillation frequencies, reduced damping, increased harmonic distortion, and additional higher-frequency oscillation modes due to the AI training load pattern.

IV. CONCLUSIONS AND FUTURE WORK

This paper presented a grid-integrated electromagnetic transient (EMT) modeling framework to analyze the impact of large-scale data center loads on power system dynamics. The model incorporates major data center subsystems, including PSUs, PDUs, UPS systems, cooling loads, and IT loads, enabling a detailed representation of converter-driven load behavior. Simulation studies on the Kundur two-Area benchmark system under three disturbance scenarios, temporary voltage drop, grid-to-UPS transition with load loss, and UPS-to-grid reconnection, show that AI training load profiles introduce additional higher-frequency oscillatory modes and slightly reduce damping ratios under certain disturbances. Frequency deviations remain within acceptable limits across all events, with the UPS transition providing rapid stabilization at the PCC. The results also indicate increased harmonic distortion compared with the baseline Kundur system due to converter-based load characteristics. Overall, while the fundamental electromechanical dynamics remain intact, large data center

loads introduce additional oscillatory interactions and power quality impacts that should be considered in grid integration studies.

Future work will focus on extending the proposed EMT modeling framework to larger interconnected power systems with multiple large-scale data centers and on developing coordinated control strategies for devices and power electronic interfaces to enhance oscillation damping.

ACKNOWLEDGMENT

This work was partly supported by the Electrical and Computer Engineering Department at UNC Charlotte, and NRECA.

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